

3.1 Managing Costs

In this course, I will:

- Start by covering when to use various cost savings strategies and how to work with the AWS Compute Optimizer tool
- Configure the S3 Requester Pays option and work with EC2 Reserved Instances as well as Spot Instances
- Tag resources for cost allocation purposes using the console, the CLI, and PowerShell
- Analyze AWS charges using the Cost Explorer tool
- Create cost and usage budgets to track monthly expenditures and also to be alerted when a certain threshold cost is reached

AWS Cost Management Strategies

AWS Cost Management

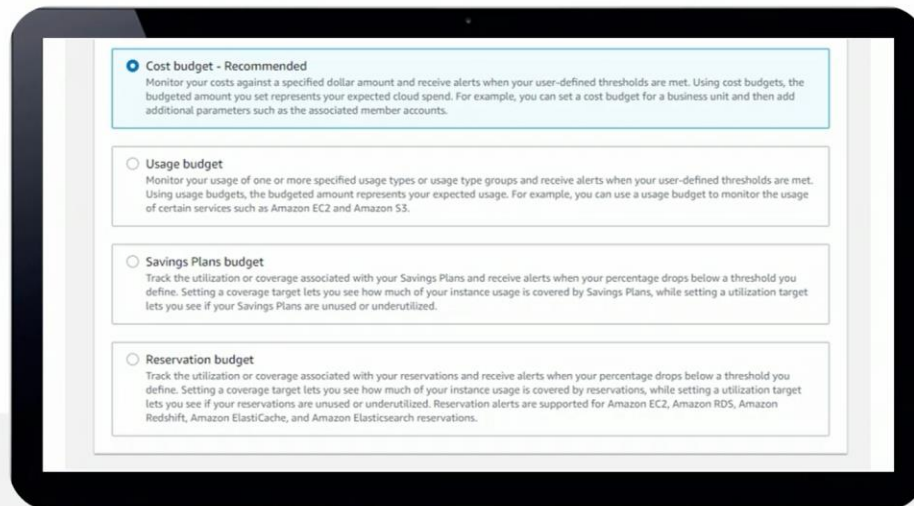
- Spot instances
- Reserved instances vs. Savings Plan
- Monitoring for usage, unused resource removal
- Monitoring for performance, instance type resizing

AWS Instance Scheduler

- Configure EC2 and RDS start and stop schedules
- Tags can be used to associate EC2 and RDS instances with schedulers
- Instances can enter hibernation mode

AWS Pricing Calculator

Creating an AWS Budget



Using AWS Compute Optimizer

Enabling S3 Bucket Requester Pays

Configuring Spot Instances

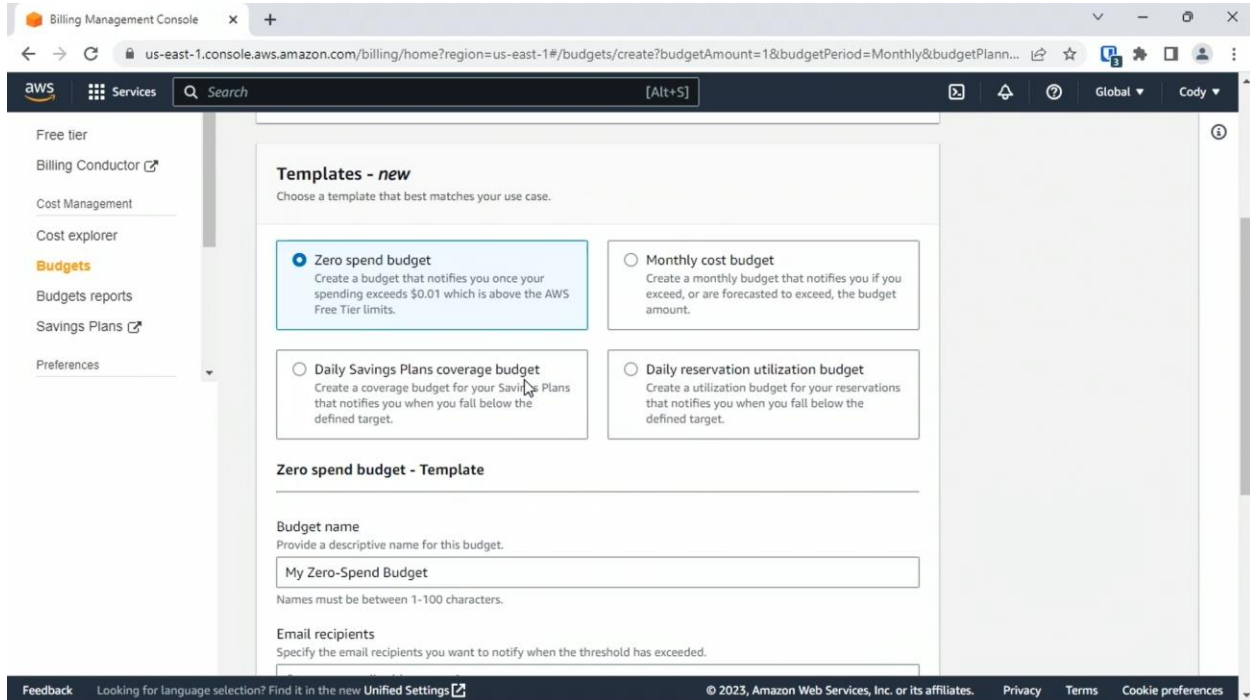
Configuring Reserved Instances

A screenshot of the AWS Management Console showing the 'Purchase Reserved Instances' dialog box. The dialog box has several filters: Platform (Windows with SQL Server S...), Availability Zone (Any), Tenancy (Default), Offering class (Standard), Instance type (t3.micro), Term (1 month to 12 months), and Payment option (All upfront). A 'Search' button is present. Below the filters, there is a table with columns: Seller, Term, Effective rate, Upfront price, Hourly rate, Availability Zone, Payment option, Offering class, and Quantity available. The table is currently empty, and a message states: 'No Reserved Instances found. There is no Reserved Instance offering that matches the search criteria you provided. Please adjust your search filters and try searching again.' The footer of the console shows 'You currently have no items in your cart.' and a 'View cart' button.

Managing Cost Allocation Tags

Analyzing AWS Costs

Creating Cost Budgets



Creating Usage Budgets

Using AWS Trusted Advisor Recommendations

1. What is the maximum number tags an AWS resource can have?
 - **50**
2. Which existing EC2 instance attributes must match your reserved instance configuration?
 - **Instance type**
 - **Region**

3. You would like to be notified when EC2 monthly charges exceed \$200.00. What should you configure?
 - **A cost budget with an EC2 filtered scope**
4. You are using the AWS Cost Explorer tool. Which file format is supported for downloading data from this page?
 - **CSV**
5. You would like to reduce monthly S3 storage costs, so you decide to configure the S3 bucket “requestor pays” option. What will the monthly S3 cost result be?
 - **Monthly costs will not be reduced due to this setting; the bucket owner pays for data storage**
6. When downloading Advisor security recommendations, which indicator implies a medium severity?
 - **Yellow**
7. What is one potential drawback of using spot instances?
 - **Workload interruption**
8. You would like to be notified when KMS request usage exceeds 1,000. What should you configure?
 - **A usage budget for KMS**
9. How do Savings Plans differ from Reserved Instances?
 - **Savings Plans are based on committing to an hourly spend value**
 - **Savings Plans apply to all instance types in all regions**
10. Over which time frame does the AWS Compute Optimizer track resource usage in order to generate recommendations?
 - **14**